

ECON 712: Applied Econometrics

Lecture 15: Panel Data: Introduction

Motivation: Panel Data and Unobserved Heterogeneity

The Within (Fixed Effects) Estimator

Why Panel Data?

- A **panel dataset** observes the same n units (firms, workers, countries) across T time periods
- Cross-sectional OLS fails when the regressor of interest is correlated with **unobserved, time-invariant** unit characteristics α_i
- Example: estimating the effect of capital on output when unobserved managerial quality α_i is correlated with capital x_{it} and output y_{it}
- With panel data, we can control for α_i *without measuring it* — by **subtracting it away** using within-unit variation over time
- Mixtape (Cunningham, 2021): “Fixed effects is about removing the omitted variable bias caused by unobserved, time-invariant heterogeneity”

Long format (stored in Stata/R):

i	t	y_{it}	x_{it}
1	1	y_{11}	x_{11}
1	2	y_{12}	x_{12}
$\vdots$	$\vdots$	$\vdots$	$\vdots$
1	T	y_{1T}	x_{1T}
2	1	y_{21}	x_{21}
$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	T	y_{nT}	x_{nT}

Matrix view (stacked over all NT obs):

$$\underbrace{\mathbf{y}}_{NT \times 1} = \underbrace{\mathbf{X}}_{NT \times K} \boldsymbol{\beta} + \boldsymbol{\alpha} + \boldsymbol{\varepsilon}$$

where K is the number of exogenous regressors

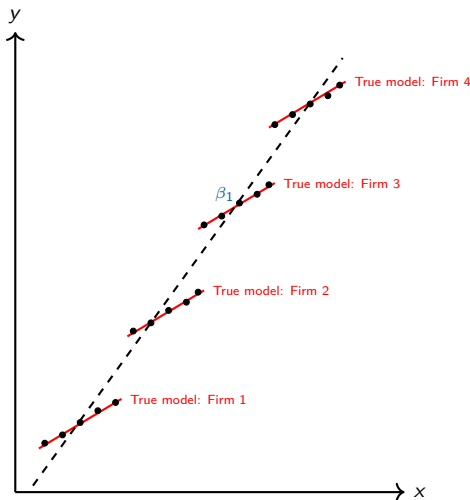
- $\mathbf{y}$: ($NT \times 1$) outcome vector, stacked unit-by-unit
- $\mathbf{X}$: ($NT \times K$) matrix of exogenous regressors
- $\boldsymbol{\alpha}$: unit fixed effects, eliminated via de-meaning
- **Balanced panel**: all NT observations present

Heterogeneity Bias: Why Pooled OLS Fails

- True model (with α_i in the error):

$$y_{it} = \beta_1 x_{it} + \underbrace{\alpha_i + \varepsilon_{it}}_{u_{it}}$$

- Each firm has the **same slope** β_1 but a different intercept α_i
- Firms with higher x also have higher $\alpha_i \Rightarrow \text{Cov}(x_{it}, \alpha_i) > 0$
- Pooled OLS (- - -) conflates **between-firm** and **within-firm** variation $\Rightarrow$ **upward bias**
- FE uses only *within-firm* time variation: recovers true β_1



Panel Data Model

- Observe $(y_{it}, \mathbf{x}_{it})$ for $i = 1, \dots, n$ and $t = 1, \dots, T$
- The workhorse **fixed effects** model is:

$$y_{it} = \mathbf{x}_{it}'\boldsymbol{\beta} + \alpha_i + \varepsilon_{it}$$

where

- ▶ α_i : **unit fixed effect** — captures all time-invariant differences across units (ability, culture, geography, ...)
 - ▶ ε_{it} : idiosyncratic error, $\mathbb{E}[\varepsilon_{it} | \mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}, \alpha_i] = 0$ (strict exogeneity)
- Key: α_i may be **arbitrarily correlated** with $\mathbf{x}_{it}$ — this is what distinguishes FE from pooled OLS

The Omitted Variable Interpretation

- Suppose the true model is $y_{it} = \mathbf{x}_{it}'\beta + \alpha_i + \varepsilon_{it}$
- Pooled OLS treats α_i as part of the composite error: $u_{it} = \alpha_i + \varepsilon_{it}$
- Pooled OLS probability limit:

$$\text{plim } \hat{\beta}^{OLS} = \beta + \underbrace{\left(\mathbb{E}[\mathbf{x}_{it}\mathbf{x}_{it}'] \right)^{-1} \mathbb{E}[\mathbf{x}_{it}\alpha_i]}_{\text{heterogeneity bias}}$$

- The bias term is zero *only* if $\text{Cov}(\mathbf{x}_{it}, \alpha_i) = \mathbf{0}$
- In the graph on the previous slide: firms with higher x_{it} also have higher α_i , so $\mathbb{E}[x_{it}\alpha_i] > 0$ and pooled OLS overestimates β
- The FE estimator eliminates α_i by exploiting **within-unit** time variation, recovering consistent $\hat{\beta}_{FE}$ regardless of $\text{Cov}(\mathbf{x}_{it}, \alpha_i)$

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The Within (Fixed Effects) Estimator

De-meaning: Subtracting the Individual Average

- Starting from $y_{it} = \mathbf{x}_{it}'\beta + \alpha_i + \varepsilon_{it}$, average over time for unit i :

$$\bar{y}_i = \bar{\mathbf{x}}_i'\beta + \alpha_i + \bar{\varepsilon}_i$$

where $\bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}$, and similarly for $\bar{\mathbf{x}}_i$, $\bar{\varepsilon}_i$

- Subtract the time-mean from each observation (**within transformation**):

$$\underbrace{y_{it} - \bar{y}_i}_{\tilde{y}_{it}} = \underbrace{(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)'}_{\tilde{\mathbf{x}}_{it}'} \beta + \underbrace{(\varepsilon_{it} - \bar{\varepsilon}_i)}_{\tilde{\varepsilon}_{it}}$$

- α_i **drops out exactly** because it is time-invariant: $\alpha_i - \bar{\alpha}_i = 0$
- Strict exogeneity implies $\mathbb{E}[\tilde{\mathbf{x}}_{it}\tilde{\varepsilon}_{it}] = \mathbf{0}$, so OLS on the demeaned model is consistent

The FE Estimator

- The **Fixed Effects (Within)** estimator is OLS on the de-meanned data:

$$\hat{\beta}_{FE} = \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{\mathbf{x}}_{it} \ddot{\mathbf{x}}'_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{\mathbf{x}}_{it} \ddot{y}_{it} \right)$$

- Equivalently, in matrix notation, define the within-group projector $M_i = I_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}'_T$ for each unit block. Then $\hat{\beta}_{FE} = (\ddot{\mathbf{X}}' \ddot{\mathbf{X}})^{-1} \ddot{\mathbf{X}}' \ddot{\mathbf{y}}$
- Consistency: as $N \rightarrow \infty$ with T fixed,

$$\hat{\beta}_{FE} \xrightarrow{p} \beta$$

provided strict exogeneity holds and $\text{rank}(\sum_{i,t} \ddot{\mathbf{x}}_{it} \ddot{\mathbf{x}}'_{it}) = k$

- Time-invariant regressors (z_i : race, gender, region) cannot be identified — they are collinear with α_i

- An equivalent formulation is the **Least Squares Dummy Variables (LSDV)** estimator:

$$y_{it} = \mathbf{x}_{it}'\beta + \sum_{j=1}^N \alpha_j \mathbf{1}[i = j] + \varepsilon_{it}$$

i.e., include a full set of unit dummies $d_{1i}, \dots, d_{Ni}$

- By the **Frisch–Waugh–Lovell Theorem**, the OLS estimate of β in LSDV equals $\hat{\beta}_{FE}$:
 - ▶ Partial out unit dummies from $\mathbf{y}$ and $\mathbf{X} \Rightarrow$ residuals are $\check{y}_{it}$ and $\check{\mathbf{x}}_{it}$
 - ▶ OLS on the residuals gives the within estimator
- LSDV and de-meaning are *numerically identical* for $\hat{\beta}$
- Practical note: LSDV with large n creates $n - 1$ dummies– this requires a lot of RAM; de-meaning is computationally faster (e.g., `areg` or `reghdfe` in Stata)

Panel Data: Additional Details

- If using demeaned variables, the degrees of freedom needs to be adjusted ($nT - n - k$)— *reghdfe* in STATA adjusts the degree of freedoms calculation automatically
- *One-way* FE models control for individual time-invariant characteristics that are correlated with the exogenous variable, but it does not control for time-variant characteristics affecting all observations across time

- ▶ To control for this we need to use *two-way* FE:

$$y_{it} = \mathbf{x}_{it}'\beta + \alpha_i + \lambda_t + \varepsilon_{it}$$

- ▶ Identification, unlike in the *one-way* FE case, is given by relative changes across time
- ▶ This is directly related to *differences-in-differences*, which is the last topic of the semester